

Ali Younis

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Machine Learning and Computer Vision engineer specializing in robotics and 3D perception with extensive experience in developing practical ML/CV systems, deployable ML, embedded devices and general software development.

Education

Ph.D. Computer Science (Machine Learning & Computer Vision)	University of California, Irvine
Thesis[1] Focus: Learnable particle systems for computer vision. Advisor: Erik Sudderth	<i>Oct. 2020 – June 2025</i>
M.S. Computer Science	University of California, Irvine
Focus: Machine Learning and Artificial Intelligence GPA: 3.882	<i>Sept. 2017 – June 2018</i>
B.S. Computer Science and Engineering	University of California, Irvine
Dean's Honor List (All Quarters) Magna Cum Laude GPA: 3.851	<i>Oct. 2013 – June 2017</i>

Experience

Senior Machine Learning and Computer Vision Engineer - OpenSpace	<i>June 2025 – Present</i>
- Developing and deploying 3D reconstruction and 3D scene understanding systems using only 2D video data in challenging conditions: motion blur, poor lighting, dirty/damaged lens. Using state-of-the-art customized feed-forward 3D reconstruction models (MapAnything, VGGT, etc.) and World Models. - Developing and deploying custom blurry and degraded image correction models for use in 3D reconstruction. - Within first 4 months, shipped critical, customer facing updates to existing indoor localization systems, which use only passive sensors (no beacons), improving the accuracy of the system significantly (upwards of 30%). - Within first 6 weeks, developed and deployed a novel neural 360 image leveler, saving the team 10+ hours per week of manual correction work. - Built 3D Gaussian Splatting systems for large indoor environments. - Investigating/using Vision-Language Models (VLMs) for 2D/3D vision understanding, including how ideas from Vision-Language-Action (VLA) models may support data-driven reasoning, interpretation, and decision-making.	

Graduate Student Researcher - University of California, Irvine	<i>Oct. 2020 – June 2025</i>
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Thesis [1] Focus: Computer vision for perception, autonomous driving, general robotics, object tracking, etc. Creating end-to-end learnable particle-based state estimation systems which utilize the latest in neural architectures within Bayesian frameworks for superior accuracy and inference speed on multi-modal time-series data.

Notable Works:

- Differentiable Particle Belief Propagation [1]:** End-to-end learnable belief propagation systems for 3D human mesh reconstruction from video. Combined learnable networks with novel end-to-end differentiable particle belief propagation to improve reliability of 3D human skeleton prediction while modeling uncertainty accurately.
- Differentiable Particle Filters [3]:** Created first end-to-end differentiable particle filter algorithm with unbiased gradients for computer vision tasks on time-varying (time series) systems with multi-modal data. Application: Real-time localization in unseen environments using only RGB camera data and odometry information. Accurately estimates multiple modes while modeling uncertainty.
- Differentiable Particle Smoothers [2]:** Created the first end-to-end differentiable particle smoother for computer vision tasks on time-varying (time series) systems with multi-modal data. Used bird's-eye view (BEV), RGB cameras and vehicle odometry for real-time vehicle localization in large real-world city-scale environments. Achieved 2x accuracy and 3x performance improvements over existing state-of-the-art methods.
- Path Planning:** Developed novel particle based path planning algorithms for planning in high dimensional and complex environments using graphical models.

Software Engineering Intern (Machine Learning, PhD) - Qualcomm Inc.	<i>June 2024 – Sept. 2024</i>
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Computer Vision Algorithms Research Group. Developed a novel machine learning based depth-from-stereo method based on differentiable non-parametric loopy belief propagation, trained end-to-end with neural networks tailored for mobile deployment. Assisted in commercialization (mobile deployment) of past work. Implemented training and evaluation pipelines in PyTorch.

Software Engineering Intern (Machine Learning, PhD) - Qualcomm Inc.	<i>June 2023 – Sept. 2023</i>
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Computer Vision Algorithms Research Group. Developed a proprietary depth-from-stereo method which blends classical and neural computer vision techniques while leveraging existing custom on-chip hardware accelerators. Method significantly outperformed existing real-time stereo methods (1.5x accuracy improvement). Implemented training and evaluation pipelines in PyTorch.

Software Engineer / Machine Learning Engineer (Part time) - Modal AI

June 2020 – June 2023

Developed, trained and deployed (to embedded) real-time depth-from-stereo algorithms (classic and machine learning based) for applications in mapping and collision avoidance for drones. Algorithm development for GPU using OpenCL. Implemented data collection, training, validation, model quantization and deployment pipelines. Integrated various Visual Inertial Odometry systems and created custom camera drivers. Android development. Built custom Linux images.

Software Engineer - Tyvak Nano-Satellite Systems, Inc.

Aug. 2018 – June 2020

Designed and implemented mission critical embedded software systems for satellites. Engineer responsible for core software libraries and several spacecraft subsystems (sensors, actuators, control). **Software lead** for all computer vision based camera payloads and for mission with first consumer-grade GPU in space. Platform bring up and Linux OS customization. Continuous integration maintainer. Member of satellite early stage (first days in orbit) launch operations team, debugging issues in real time. Facilitated inter-team cooperation by holding conflict resolution/sync-up meetings for various high stakes missions.

Software Engineering Intern - Qualcomm Inc. Corporate R&D

4 Summers from 2014 – 2017

Designed and deployed static/dynamic object collision avoidance system for drones using only on-board sensors (RGB/RGB-D/IMU) and on-board processing; 3D mapping, state estimation and path planning (prototype for 2016 CES demo). Implemented customer facing UART driver. Created continuous integration testing tools. Developed new features for the drone flight stack; safety, GPS navigation, ground station communications and visualizations. Received several excellent teamwork awards as well as a scholarship for outstanding performance.

Undergraduate Research Asst. - University of California, Irvine

Nov. 2013 – June 2017

Researcher in the Programming Language Research Group under Professor Brian Demsky. Developed a decentralized secure communications system for IoT devices [5]. Researched security of IoT devices [6]. Created a bare-metal ARMv7 test platform for code on "unreliable" hardware research. Implemented lock-free concurrent data structures in C++11 using weak memory models. Wrote device drivers and Android applications.

Skills

- **Languages** C, C++, Java, Python, Go, Bash, MATLAB, LaTeX, Node.js, SQL, ARMv7 Assembly
- **Frameworks/Tools:** PyTorch, TensorFlow, 3D Gaussian Splats, GTSAM, Keras, TFLite, ONNX, CUDA, OpenCL, OpenCV, Make, CMake, Buildroot, Linux, Android, iOS, Google Cloud, MySQL, Postgres, SLURM, LSF
- **Neural Architectures:** Vision-Language Models (VLMs), Vision-Language-Action Models (VLAs), Vision Transformers (ViT), Foundation World Models, Feed-Forward 3D Reconstruction Models (VGGT, MapAnything, DepthAnythingV3, etc.), CNNs, RNNs, Transformers, Spatial Transformers, Segment Anything (SAM), DinoV2/V3, YOLO, Mono/Stereo Depth Networks, ResNets, UNets, Diffusion Models.

Publications

- [1] **A. Younis**. "End-to-end Differentiable Learning of Particle Belief Propagation Algorithms". Doctoral Thesis, University of California, Irvine. June 2025.
- [2] **A. Younis**, E. Sudderth. "Learning to be Smooth: An End-to-End Differentiable Particle Smoother". Neural Information Processing Systems (NeurIPS) 2024.
- [3] **A. Younis**, E. Sudderth. "Differentiable and Stable Long-Range Tracking of Multiple Posterior Modes". Neural Information Processing Systems (NeurIPS) 2023.
- [4] S. Agarwal, G. Hope, **A. Younis**, E. Sudderth. "A Decoder Suffices for Query-Adaptive Variational Inference". Uncertainty in Artificial Intelligence (UAI) 2023.
- [5] R. Trimananda, **A. Younis**, T. Kwa, B. Demsky, G. Xu. "Securing Smart Home Devices against Compromised Cloud Servers". Poster at HotEdge, June 2020.
- [6] R. Trimananda, **A. Younis**, B. Wang, B. Xu, B. Demsky, G. Xu. "Vigilia: Securing smart home edge computing". Symposium on Edge Computing (SEC) October 2018.

Other Experience

- Reviewer at NeurIPS 2024.
- In charge of research lab's compute cluster (administration, upgrading hardware, coordination).
- Teaching assistant for various Machine Learning Courses: CS 178, CS 275P.